



Submitted By :	Abdul Haseeb
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Topic:	Node JS Installation
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Objectives

- Node JS installation
- Working with Node JS. Sending an email with Node JS

Q1: Install Node.js on your machine by following above manual

Open your browser and go to the official Node.js website:

<https://nodejs.org>

we will see two versions:

- **LTS (Long Term Support)**
- **Current**

Choose LTS if you're just starting out.

The screenshot shows the 'Download Node.js®' page. At the top, there are dropdown menus to select the version (v22.20.0 LTS), operating system (Windows), installation method (Docker), and package manager (npm). Below this is a green banner with an 'Info' icon and text: 'Want new features sooner? Get the latest Node.js version instead and try the latest improvements!'. The main content is a dark-themed code block with white text showing Docker commands for installing Node.js. The commands are numbered 1 through 14. At the bottom of the code block, there is a 'PowerShell' label and a 'Copy to clipboard' button. Below the code block, there is a note: 'Docker is a containerization platform. If you encounter any issues please visit Docker's website'. At the bottom of the page, there are more dropdown menus to select the operating system (Windows) and architecture (x64), followed by two green buttons: 'Windows Installer (.msi)' and 'Standalone Binary (.zip)'.

```
1 # Docker has specific installation instructions for each operating system.
2 # Please refer to the official documentation at https://docker.com/get-started/
3
4 # Pull the Node.js Docker image:
5 docker pull node:22-alpine
6
7 # Create a Node.js container and start a Shell session:
8 docker run -it --rm --entrypoint sh node:22-alpine
9
10 # Verify the Node.js version:
11 node -v # Should print "v22.20.0".
12
13 # Verify npm version:
14 npm -v # Should print "10.9.3".
```

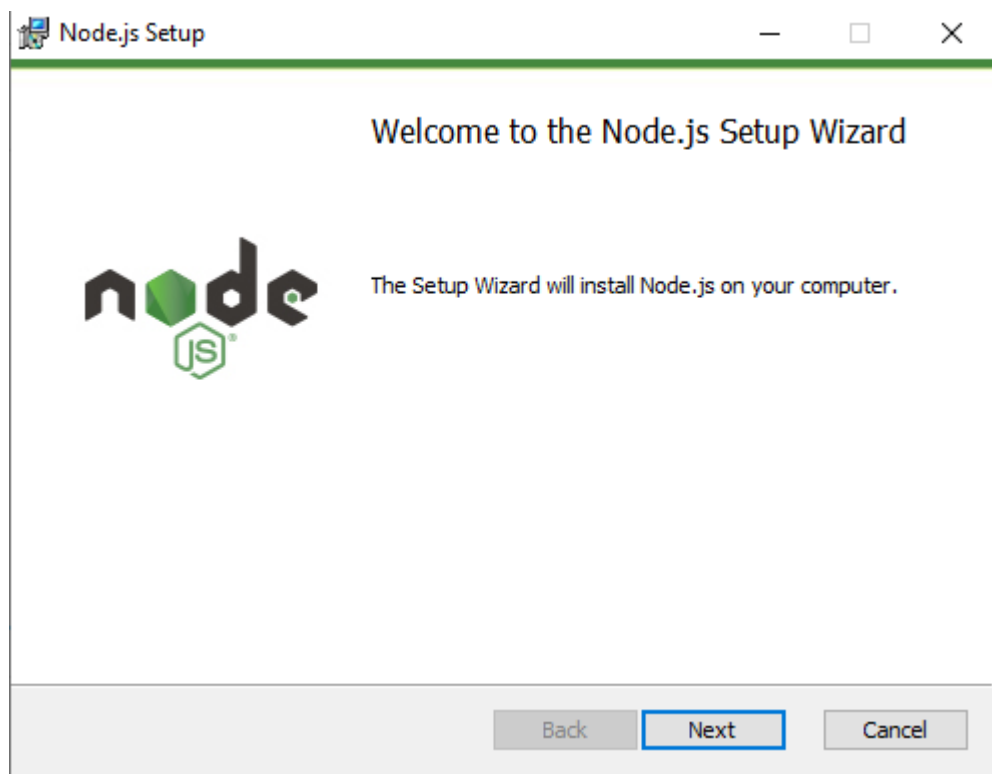
PowerShell [Copy to clipboard](#)

Docker is a containerization platform. If you encounter any issues please visit [Docker's website](#)

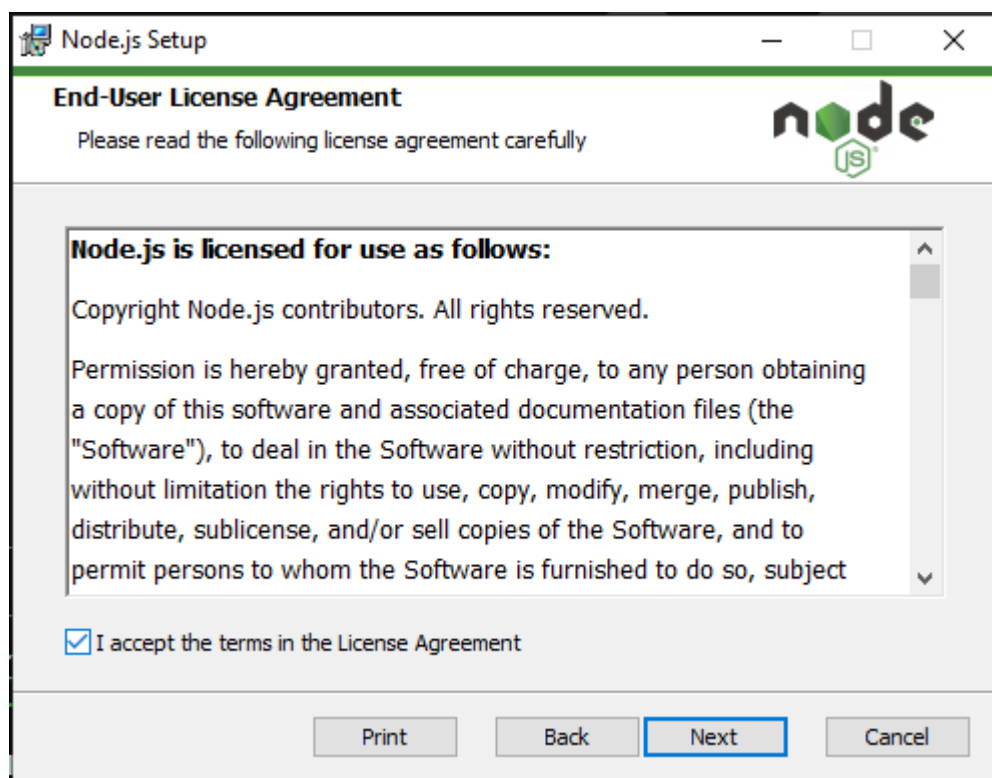
Or get a prebuilt Node.js® for Windows running a x64 architecture.

[Windows Installer \(.msi\)](#) [Standalone Binary \(.zip\)](#)

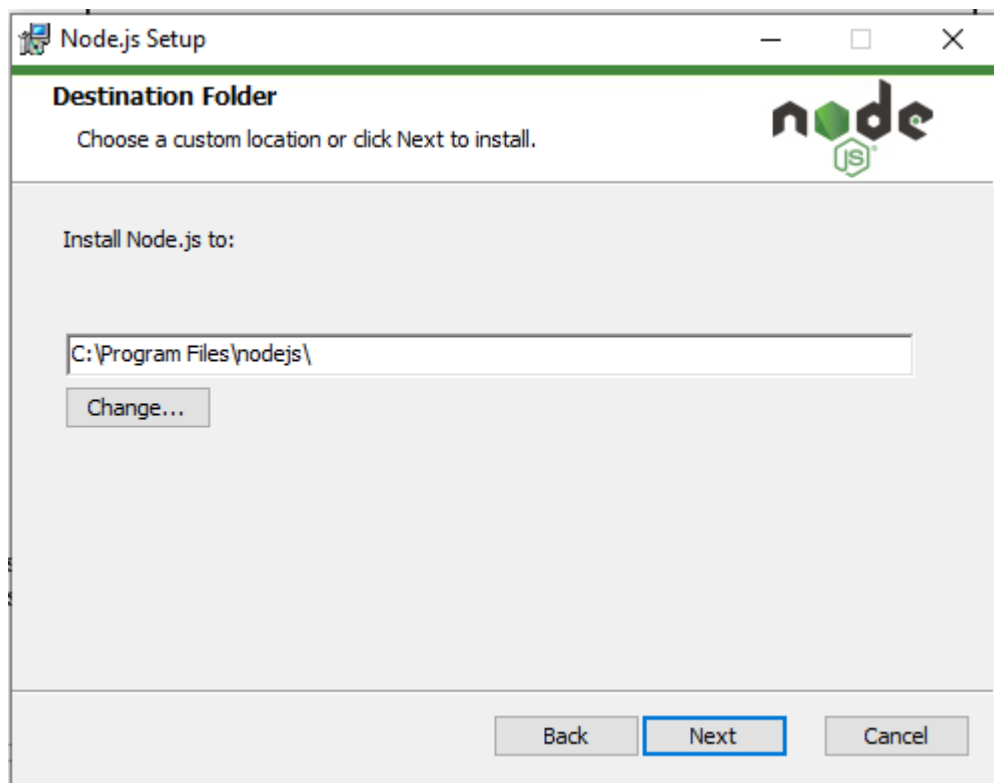
Download the .msi installer for Windows. Double-click the downloaded file.



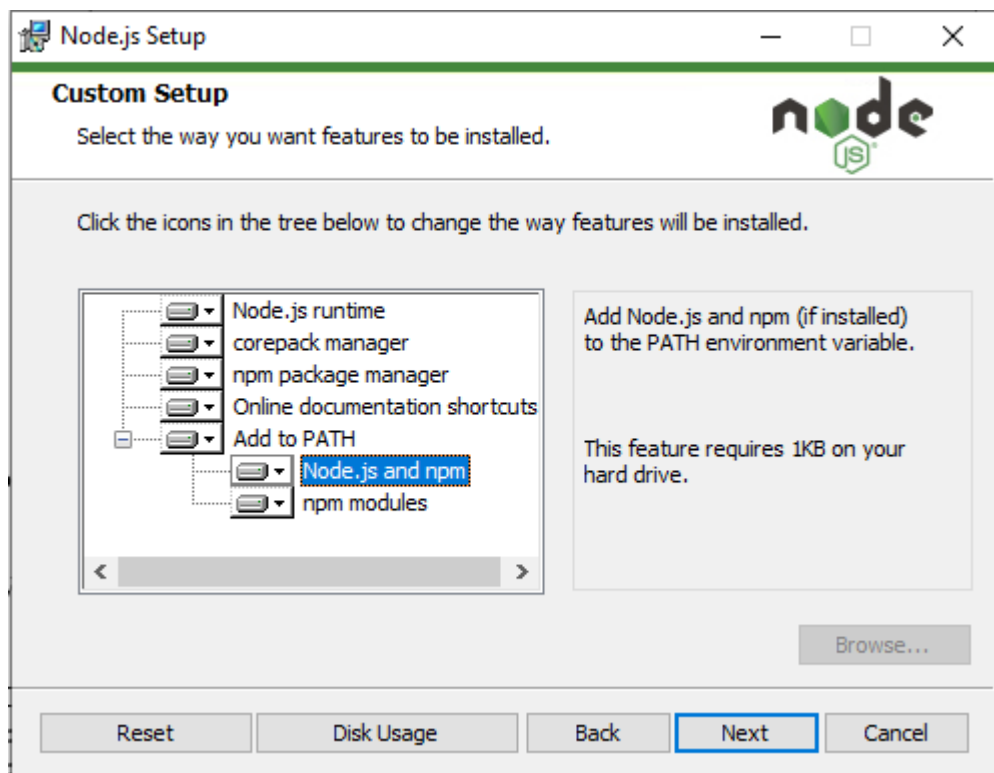
Follow the setup wizard. Accept the license agreement

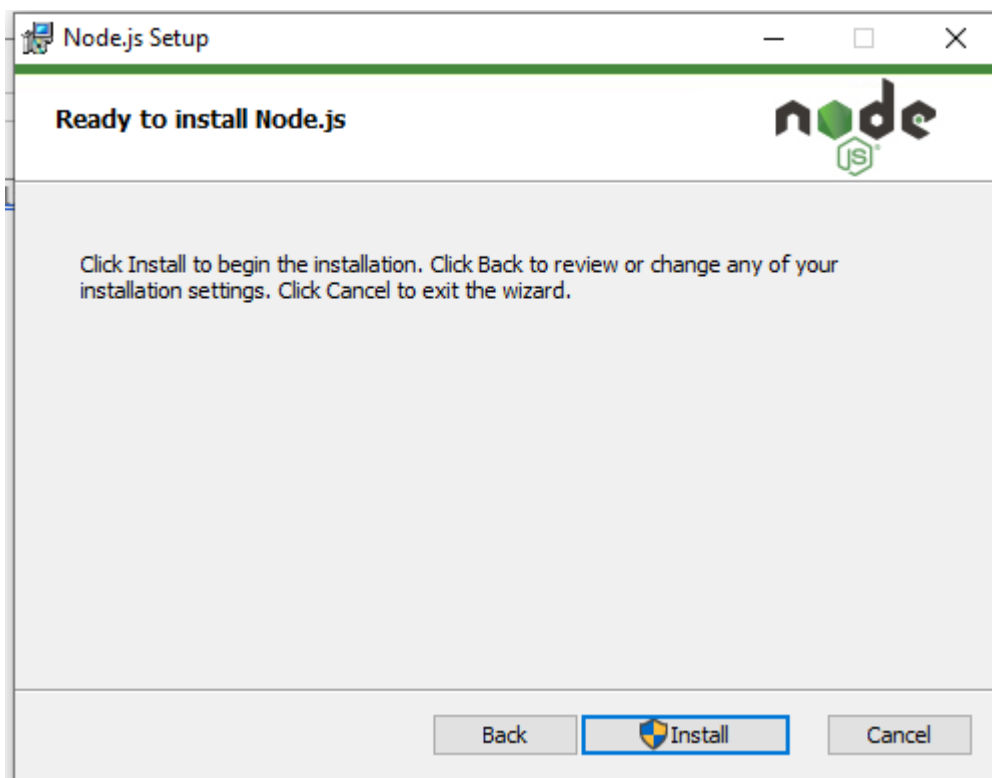
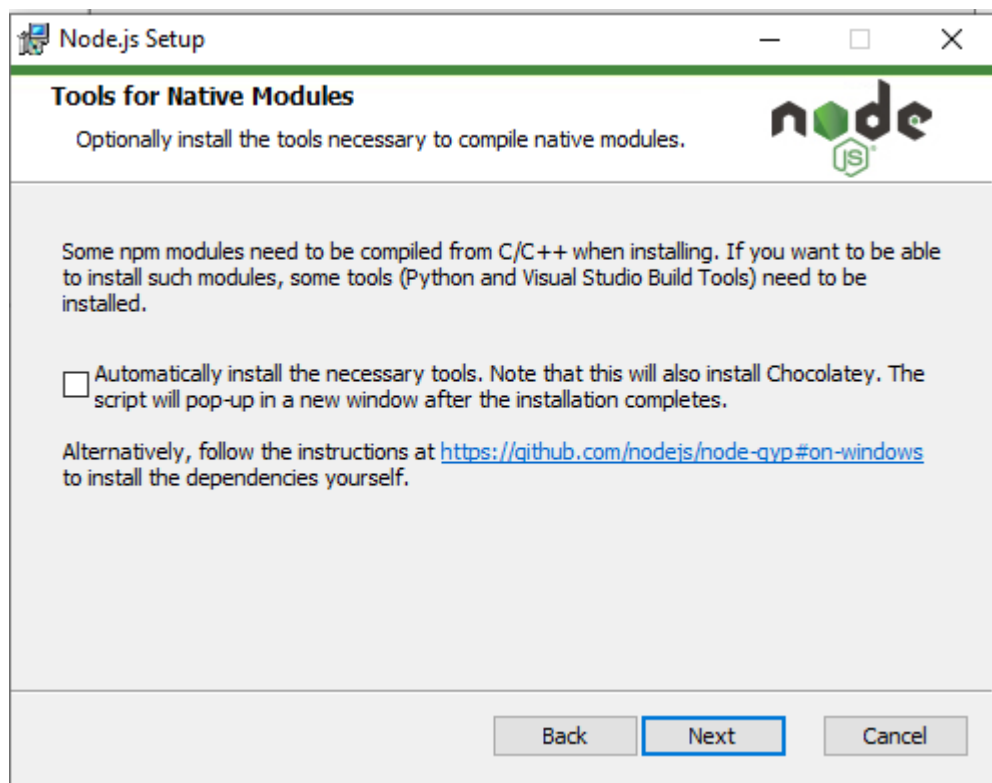


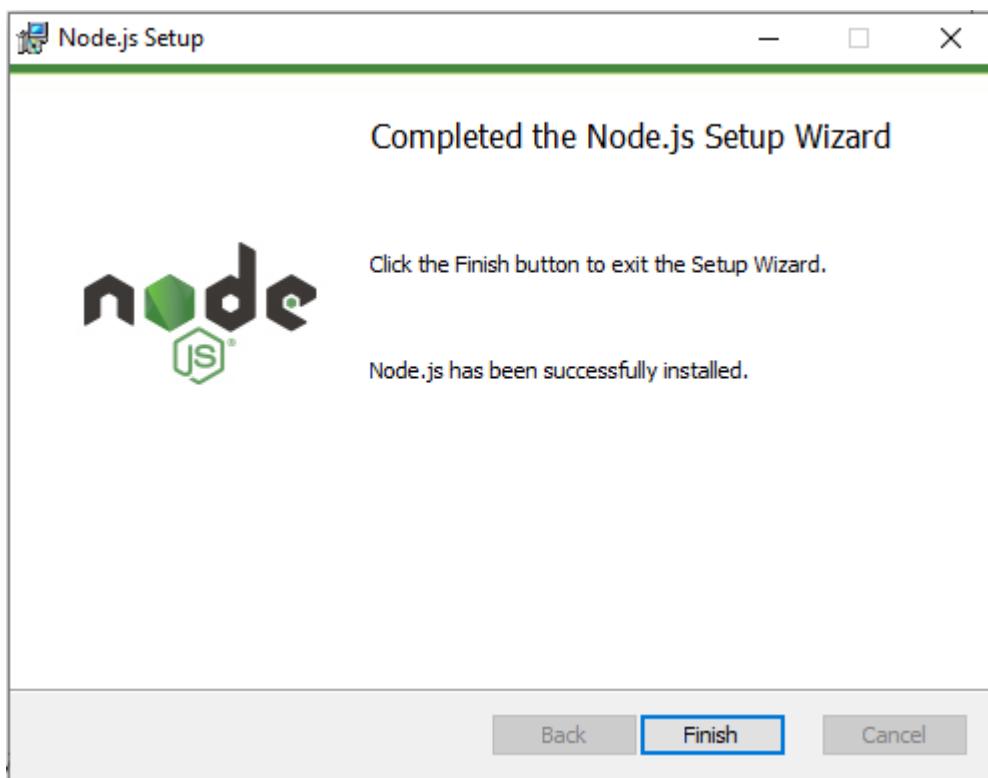
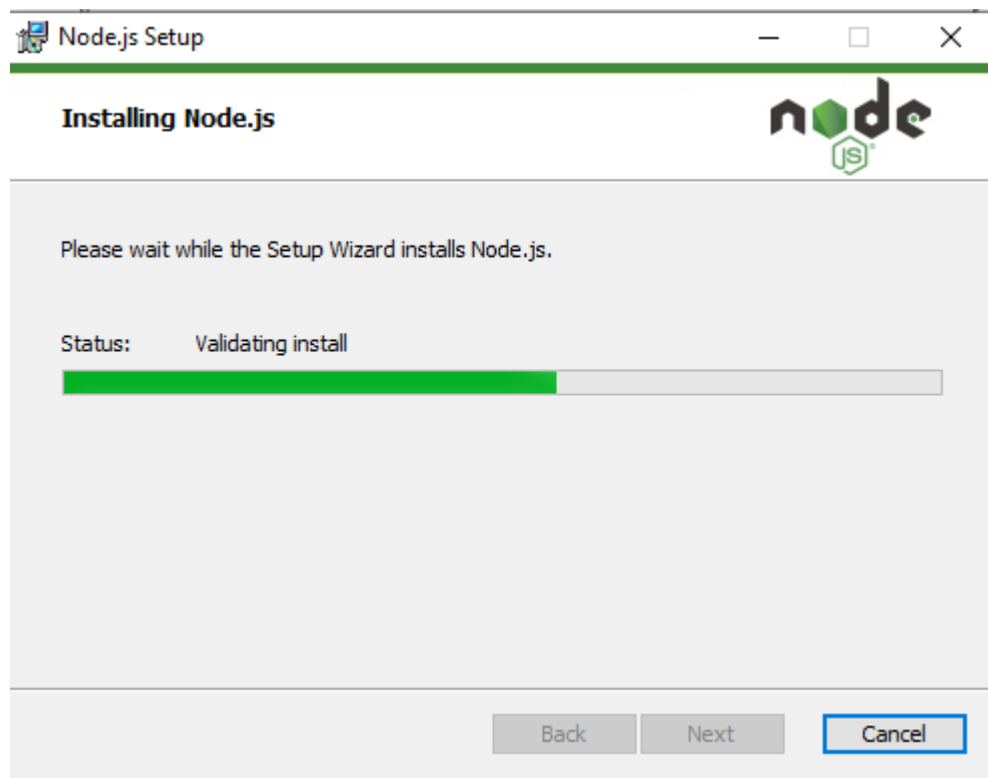
Choose the installation folder (default is fine).



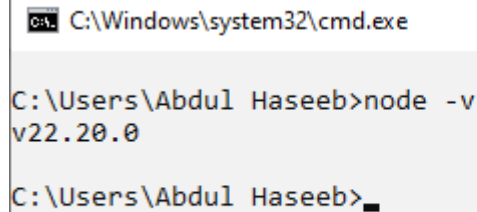
Keep “Add to PATH” checked (important). Click **Next** → **Install**.







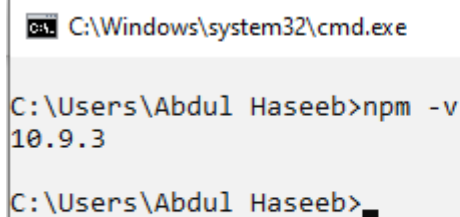
Q2: Verify installation and prepare lab manual

node -v

```
C:\Windows\system32\cmd.exe

C:\Users\Abdul Haseeb>node -v
v22.20.0

C:\Users\Abdul Haseeb>_
```

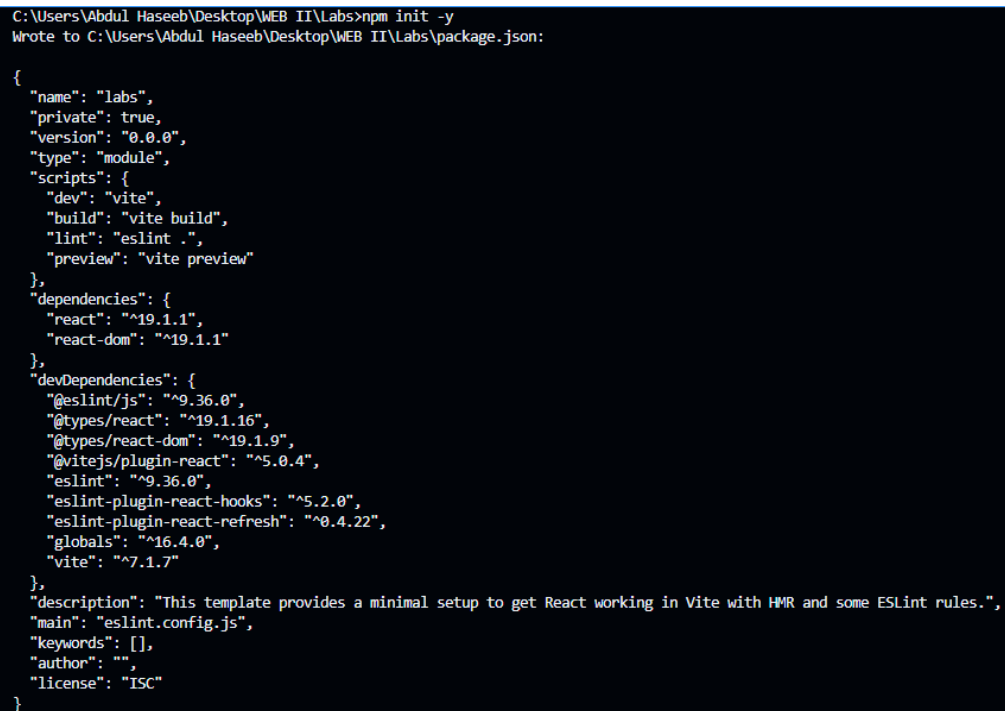
npm -v

```
C:\Windows\system32\cmd.exe

C:\Users\Abdul Haseeb>npm -v
10.9.3

C:\Users\Abdul Haseeb>_
```

Q3: Create an app to send Email from Node.js. Follow the tutorial on:
https://www.w3schools.com/nodejs/nodejs_email.asp For reference:
<https://blog.logrocket.com/send-emails-nodejs-nodemailer/>

Initialize a Node.js Project

```
C:\Users\Abdul Haseeb\Desktop\WEB II\Labs>npm init -y
Wrote to C:\Users\Abdul Haseeb\Desktop\WEB II\Labs\package.json:

{
  "name": "labs",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "react": "^19.1.1",
    "react-dom": "^19.1.1"
  },
  "devDependencies": {
    "@eslint/js": "^9.36.0",
    "@types/react": "^19.1.16",
    "@types/react-dom": "^19.1.9",
    "@vitejs/plugin-react": "^5.0.4",
    "eslint": "^9.36.0",
    "eslint-plugin-react-hooks": "^5.2.0",
    "eslint-plugin-react-refresh": "^0.4.22",
    "globals": "^16.4.0",
    "vite": "^7.1.7"
  },
  "description": "This template provides a minimal setup to get React working in Vite with HMR and some ESLint rules.",
  "main": "eslint.config.js",
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Install Nodemailer

```
C:\Users\Abdul Haseeb\Desktop\WEB II\Labs\backend>npm install nodemailer

added 1 package, and audited 2 packages in 2s

found 0 vulnerabilities
```

Create server.js

```
const express = require("express")
const nodemailer = require("nodemailer")
const cors = require("cors")

const app = express()
app.use(cors())
app.use(express.json())

app.post("/send-email", async (req, res) => {
  const { email, message } = req.body

  try {
    let testAccount = await nodemailer.createTestAccount()

    let transporter = nodemailer.createTransport({
      host: "smtp.ethereal.email",
      port: 587,
      secure: false,
      auth: {
        user: "lazaroschmeller@ethereal.email",
        pass: "ebWpywDBN4tNqV3QcW",
      },
    })

    let info = await transporter.sendMail({
      from: '"Abdul Haseeb" <no-reply@example.com>',
      to: email,
      subject: "Message from Haseeb",
      text: message,
      html: `<p>${message}</p>`,
    })

    console.log("Preview URL: " + nodemailer.getTestMessageUrl(info))

    res.json({
```

```
        success: true,
        previewURL: nodemailer.getTestMessageUrl(info),
    })
} catch (err) {
    console.error(err)
    res.json({ success: false, error: "Email not sent" })
}
})

const PORT = 5000
app.listen(PORT, () => console.log(` Server running on port ${PORT}`))
```

SendEmail.js

```
import { useState } from "react"

function SendEmail() {
    const [email, setEmail] = useState("")
    const [message, setMessage] = useState("")
    const [response, setResponse] = useState("")

    const handleSubmit = async (e) => {
        e.preventDefault()

        try {
            const res = await fetch("http://localhost:5000/send-email", {
                method: "POST",
                headers: { "Content-Type": "application/json" },
                body: JSON.stringify({ email, message }),
            })

            const data = await res.json()
            if (data.success) {
                setResponse(`Email sent! Preview: ${data.previewURL}`)
            } else {
                setResponse("Failed to send email")
            }
        } catch (error) {
            setResponse("Error: " + error.message)
        }
    }

    return (
        <div style={{ padding: "20px", maxWidth: "400px", margin: "auto" }}>
```

```
<h2>Send Email</h2>
<form onSubmit={handleSubmit}>
  <input
    type="email"
    placeholder="Recipient Email"
    value={email}
    onChange={(e) => setEmail(e.target.value)}
    required
    style={{ width: "100%", marginBottom: "10px", padding: "8px" }}
  />
  <textarea
    placeholder="Your Message"
    value={message}
    onChange={(e) => setMessage(e.target.value)}
    required
    style={{ width: "100%", marginBottom: "10px", padding: "8px" }}
  />
  <button type="submit" style={{ padding: "10px 20px" }}>
    Send
  </button>
</form>
{response && <p>{response}</p>}
</div>
)
}

export default SendEmail
```

App.js

```
import SendEmail from "../Components/sendEmail"
```

```
function App() {
  return (
    <>
      <SendEmail />
    </>
  )
}
```

```
export default App
```

Output

Send Email

lazaroschmeler@ethereal.email

Hi my name is Abdul haseeb and i am sending this message to test fake smtp

Send

Mailbox before email received

Ethereal

Home

FAQ

Help

Messages

Logout

Messages for **lazaroschmeler@ethereal.email** – this account can not be used for inbound emails ⓘ

Page 1

← Newer

Older →

★

Ethereal Email

<info@ethereal.email>

This is your ethereal mailbox

Today at 23:38

© 2017—2022 Ethereal Email info@ethereal.email | service provided by [EmailEngine](#)

Mailbox after email received

Ethereal

Home

FAQ

Help

Messages

Logout

Messages for **lazaroschmeler@ethereal.email** – this account can not be used for inbound emails ⓘ

Page 1

← Newer

Older →

To: <lazaroschmeler@ethereal.email>

Message from Haseeb

Today at 23:50

★

Ethereal Email

<info@ethereal.email>

This is your ethereal mailbox

Today at 23:38

Ethereal

Home

FAQ

Help

Messages

Logout

Headers

Envelope

Source

[Public URL of this message](#)

Subject: Message from Haseeb

From: Abdul Haseeb <no-reply@example.com>

To: <lazaroschmeler@ethereal.email>

Time: Today at 23:50

Message-ID: <0444f51c-c6cb-260a-6e26-279918372b49@example.com>

☒ HTML ☐ Plaintext

Hi my name is Abdul haseeb and i am sending this message to test fake smtp